

Consistent Partial Least Squares: A cSEM tutorial

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Abstract

This tutorial article provides a comprehensive, step-by-step illustration of consistent partial least squares (PLSc) using the **cSEM** package in R. PLSc is a variant of partial least squares path modeling, a composite-based approach to structural equation modeling, that produces consistent parameter estimates for latent variable models. In particular, PLSc corrects construct correlations for attenuation using Dijkstra-Henseler's composite reliability coefficient ϱ_A . Building on the employee retention example introduced by Cheah, Sarstedt, Hair, and Ringle (2026), we demonstrate how to install and load **cSEM**, specify the model using **lavaan**-style syntax, estimate the PLSc-SEM model, evaluate measurement model quality (internal consistency reliability, convergent validity, and discriminant validity), and assess the structural model (collinearity, effect sizes, model fit, and path coefficients with bootstrap inference). We further contrast the PLSc-SEM results with standard PLS-SEM estimates and discuss practical considerations for using PLSc-SEM in empirical research.

Keywords: measurement error, bias correction, partial least squares structural equation modeling (PLS-SEM), factor score regression

1 Introduction

Structural equation modeling (SEM) is a versatile multivariate analysis framework used in the behavioral, social, medical, and management sciences to study complex relationships between constructs and their relations with observed variables (e.g., Kline, 2023; Marcoulides & Schumacker, 1996). To estimate structural equation models, various approaches have been proposed, including maximum-likelihood (Jöreskog, 1970), weighted least squares (Browne, 1974), factor score regression (Devlieger & Rosseel, 2017; Skrandal & Laake, 2001) and (integrated) generalized structured component analysis (Hwang et al., 2021; Hwang & Takane, 2004)

A composite-based estimator to SEM that gained popularity over the last decades in various fields of social sciences (Cook & Forzani, 2023), including marketing (Sarstedt et al., 2022), information systems research (Benitez, Henseler, Castillo, & Schubert, 2020; Evermann & Rönkkö, 2023), human resource management (Ringle, Sarstedt, Mitchell, & Gudergan, 2020), and tourism research (Müller, Schubert, & Henseler, 2018), is partial least squares path modeling (PLS-PM, Wold, 1982), also known as partial least squares structural equation modeling (PLS-SEM, Hair, Ringle, & Sarstedt, 2011). PLS-PM is a two-step estimation procedure (Schubert, Zaza, & Henseler, 2023). In the first step, construct scores are obtained by the partial least squares algorithm. Subsequently, these construct scores are used to estimate the model parameters. Although PLS-PM is computationally efficient, it is known that in its original form to produce inconsistent estimates for models involving latent variables due to attenuation bias (e.g., Dijkstra, 1985; Hui & Wold, 1982; Rönkkö & Evermann, 2013; Schubert, Rosseel, et al., 2023).

To address this limitation, consistent partial least squares (PLSc, Dijkstra, 2013; Dijkstra & Henseler, 2015a, 2015b; Dijkstra & Schermelleh-Engel, 2014) was introduced. PLSc is based on PLS-PM, but applies a correction for attenuation to obtain consistent estimates of reflective measurement models and path coefficient between latent variables. Additionally, PLSc has been extended to deal with ordinal variables (Schubert, Henseler, & Dijkstra, 2018), randomly occurring outliers (Schamberger, Schubert, Henseler, & Dijkstra, 2020) and multicollinearity issues (Jung & Park, 2018).

To apply PLS-PM and PLSc, various implementations have been developed (Venturini, Mehmetoglu, & Latan, 2023). They can be distinguished into: (i) proprietary software and (ii) open-source software. Proprietary software for conducting PLS-PM and PLSc includes SmartPLS (Ringle, Wende, & Becker, 2024), WarpPLS (Kock, 2025) and ADANCO (Henseler, 2025). On the other hand, open-source software alternatives include implementations in the statistical programming environment R (R Core Team, 2025) such as *matrixpls* (Rönkkö, 2022), *SEM-inR* (Ray, Danks, & Calero Valdez, 2026), *cSEM* (Rademaker & Schubert, 2020) and *plspm* (Sanchez, Trinchera, Russolillo, & Bertrand, 2025). Similarly, it includes implementations in python (Python Software Foundation, 2024) such as *plspm* (Humble, 2024). While proprietary software such as SmartPLS often shows high ease of use (Cheah, Magno, & Cassia, 2024), they conflict with the open science principles (UNESCO, 2021). For example, their codebase is not openly accessible of transparency (Morin et al., 2012), reproducibility (Ince, Hatton, & Graham-Cumming, 2012), and accessibility/inclusiveness (). Open-source software overcomes these limitations (Barba, 2022). Therefore, this tutorial presents the R package *cSEM* as a full-fledged open-source alternative to SmartPLS to apply PLS-PM and PLSc.

2 cSEM R Illustration

2.1 Getting started with R

To use the cSEM package, users first need to install R and an integrated development environment (IDE) such as RStudio. For the installation, we refer to the official guides for R (<https://cran.r-project.org>) and RStudio (<https://posit.co/download/rstudio-desktop>).

When RStudio is opened for the first time, it displays four panes (Figure 1). The two most important ones are the *script editor* (upper left), where code is written and saved, and the *Console* (lower left), where the code is executed and results are printed. The *Environment* pane (upper right) lists the objects currently loaded into memory, and the fourth pane (lower right) provides access to files, plots, and help pages.

Although code can be typed directly into the console, it is good practice to work in a script, since a script can be saved, shared, and re-run, which is essential for reproducible analyses. To create a new script, click **File** → **New File** → **R Script**. The typical workflow is to write code in the editor and run the selected line(s) with **Ctrl + Enter**. Any line beginning with **#** is treated as a comment: it documents what the code does and is ignored when the code runs.

```
# This is a comment and is not executed.  
1 + 1    # Code is executed; the result is printed in the Console.  
  
[1] 2
```

Before cSEM can be used, it must be installed. The cSEM package is available on CRAN, so the most recent stable version can be installed with:

```
install.packages("cSEM")
```

Installation only needs to be done once per machine (rerun this line to update the package later). Moreover, development versions can be installed from GitHub: <https://github.com/FloSchuberth/cSEM>. After installation, the package must be loaded at the start of every new R session before its functions become available:

```
library(cSEM)
```

With cSEM installed and loaded, we can now import the data and specify the model, as described in the following sections.

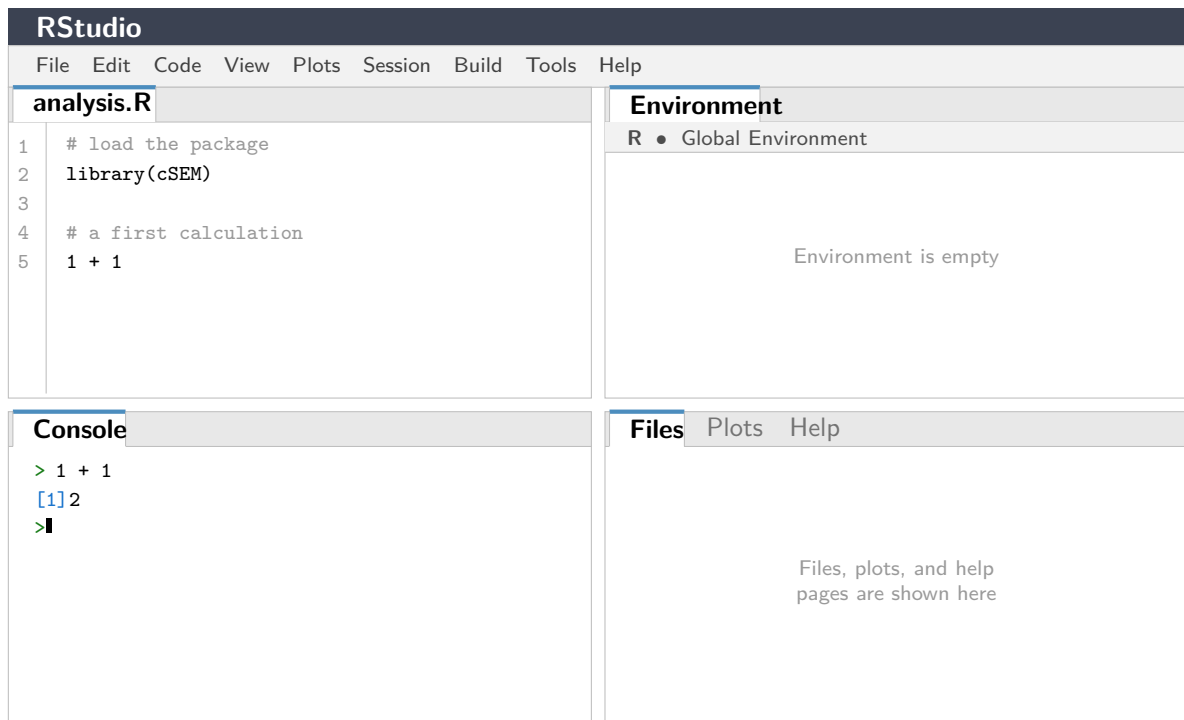


Figure 1: The four panes of the RStudio interface: the script editor (upper left), the *Console* (lower left), the *Environment* pane (upper right), and the files/plots/help pane (lower right).

2.2 Model and Data

The *cSEM* package (Rademaker & Schubert, 2020) offers a variety of composite-based approaches to SEM including PLS-PM and PLSc. Following Cheah et al. (2026), our illustration of the application of PLSc using *cSEM* draws on the employee retention model introduced by Hair, Black, Babin, and Anderson (2019, Chapter 13).

The model explains the effects of organizational commitment (OC) and job satisfaction (JS) on employees' staying intentions (SI), with work environment perceptions (EP) and attitudes toward coworkers (AC) as exogenous antecedent constructs. Five constructs are reflectively measured by a total of 21 indicators. A detailed description of the indicators can be found in Cheah et al. (2026). The model is presented in Figure 2.

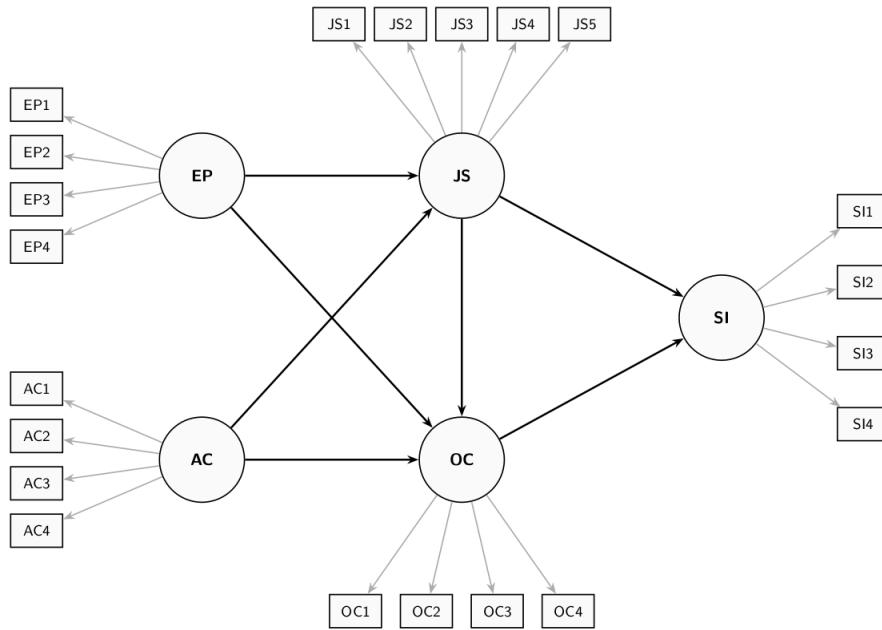


Figure 2: Employee Retention Model

The dataset used for model estimation is publicly available from the SmartPLS website: <https://www.smartpls.com/documentation/sample-projects/employee-retention>. The synthetic dataset comprises $N = 400$ observations from HBAT Industries and was generated to satisfy the assumptions of factor-based SEM, including multivariate normality. For this tutorial, we assume the data has been downloaded as a CSV file. After downloading the dataset, it needs to be loaded into the R Environment. The Employee Retention dataset is stored as an CSV File using ';' as the separator, thus it can be loaded into R using following command:

```
# Import dataset
HBAT_SEM <- read.csv("data/HBAT_SEM.csv", sep=";")

# Select relevant variables
HBAT_SEM_rel <- HBAT_SEM[, c("AC1", "AC2", "AC3", "AC4",
                             "EP1", "EP2", "EP3", "EP4",
                             "JS1", "JS2", "JS3", "JS4", "JS5",
                             "OC1", "OC2", "OC3", "OC4",
                             "SI1", "SI2", "SI3", "SI4")]

# show dimensions of the relevant dataset
dim(HBAT_SEM_rel)

[1] 400 21

# count missing values per variable
colSums(is.na(HBAT_SEM_rel))

AC1 AC2 AC3 AC4 EP1 EP2 EP3 EP4 JS1 JS2 JS3 JS4 JS5 OC1 OC2 OC3 OC4 SI1 SI2 SI3
0 0 0 1 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0
SI4
0
```

Since `cSEM` can only handle complete datasets, i.e., datasets without any missing values, we check the number of missing values per variable using the `is.na()` function. The output shows

that the variables AC4 and EP4 each contain one missing values. Before we can continue, we thus need to either apply imputation methods or delete the incomplete observations. Deleting incomplete observations can be done via following command:

```
# filter the dataset, only select complete cases
HBAT_SEM_rel <- HBAT_SEM_rel[complete.cases(HBAT_SEM_rel),]

# the number of rows changed from 400 to 398
dim(HBAT_SEM_rel)

[1] 398  21
```

Since two observations contain missing values, all analyses in the remainder of this tutorial are based on the $N = 398$ complete observations. When using their own data, users must ensure it is stored as a data frame or a matrix before passing it to **cSEM** functions; if necessary, the data can be converted with `as.data.frame()` or `as.matrix()`.

2.3 Model specification

In **cSEM**, models are specified using the **lavaan** model syntax. The **lavaan** package does not need to be installed to use the syntax. In particular, the `=~` and the `<~` operators are used to define the relations between the constructs and their indicators. Specifically, the `=~` operator is used to specify reflective measurement models, while the `<~` is used to specify composite models (see , for an explanation of reflective and composite models). In addition, the `~` operator is used to specify the relations among constructs. In case, an observed variable should be directly accounted for in the structural model, it must be specified as a single indicator construct, as common in PLS-PM (). The **cSEM** specification of the retention model illustrated in Figure 2 is as follows:

```
model <- "
  # Measurement model
  AC =~ AC1 + AC2 + AC3 + AC4
  EP =~ EP1 + EP2 + EP3 + EP4
  JS =~ JS1 + JS2 + JS3 + JS4 + JS5
  OC =~ OC1 + OC2 + OC3 + OC4
  SI =~ SI1 + SI2 + SI3 + SI4

  # Structural model
  JS ~ EP + AC
  OC ~ EP + AC + JS
  SI ~ JS + OC
"
```

3 Estimating the model with cSEM

3.1 Running the estimation

The central function is `csem()`. The minimal input required for the `csem()` function is a dataset and a model:

.data A data frame or matrix with column names matching the indicator names used in the model description. Possible column types of data proved are: "logical", "numeric" ("double" or "integer"), "factor" ("ordered" and/or "unordered"), "character" or a mix of several types.

.model A model string written in *lavaan* syntax, with `=~` for measurement relations and `~` for structural (regression) relations.

By default weights are estimated using the partial least squares path modeling algorithm ("PLS-PM").

To ultimately estimate with the `csem` function, we call it and save the results as `res_csem`:

```
# PLSc-SEM estimation
res_csem <- csem(
  .data = HBAT_SEM_rel,
  .model = model
)
```

To obtain a comprehensive overview of all results, including loadings, path coefficients, reliability measures, and R^2 values, use:

```
# Full results summary
summary_res <- summarize(res_csem)
print(summary_res)
```

```
# For a more compact overview
summary_res$Estimates>Loading_estimates
```

	Name	Construct_type	Estimate	Std_err	t_stat	p_value
1	EP =~ EP1	Common factor	0.6341011	NA	NA	NA
2	EP =~ EP2	Common factor	0.8786819	NA	NA	NA
3	EP =~ EP3	Common factor	0.7678021	NA	NA	NA
4	EP =~ EP4	Common factor	0.8230691	NA	NA	NA
5	AC =~ AC1	Common factor	0.7691299	NA	NA	NA
6	AC =~ AC2	Common factor	0.6765259	NA	NA	NA
7	AC =~ AC3	Common factor	0.8215860	NA	NA	NA
8	AC =~ AC4	Common factor	0.9933858	NA	NA	NA
9	JS =~ JS1	Common factor	0.6248337	NA	NA	NA
10	JS =~ JS2	Common factor	0.6964587	NA	NA	NA
11	JS =~ JS3	Common factor	0.7182274	NA	NA	NA
12	JS =~ JS4	Common factor	0.6319045	NA	NA	NA
13	JS =~ JS5	Common factor	0.9004003	NA	NA	NA
14	OC =~ OC1	Common factor	0.4252105	NA	NA	NA
15	OC =~ OC2	Common factor	0.9574682	NA	NA	NA
16	OC =~ OC3	Common factor	0.6858765	NA	NA	NA
17	OC =~ OC4	Common factor	0.8565670	NA	NA	NA
18	SI =~ SI1	Common factor	0.8170099	NA	NA	NA
19	SI =~ SI2	Common factor	0.8705194	NA	NA	NA
20	SI =~ SI3	Common factor	0.6778349	NA	NA	NA
21	SI =~ SI4	Common factor	0.8959341	NA	NA	NA

```
summary_res$Estimates$Path_estimates
```

	Name	Construct_type	Estimate	Std_err	t_stat	p_value
1	JS ~ EP	Common factor	0.244824960	NA	NA	NA
2	JS ~ AC	Common factor	-0.003282073	NA	NA	NA
3	OC ~ EP	Common factor	0.428781485	NA	NA	NA
4	OC ~ AC	Common factor	0.172554413	NA	NA	NA
5	OC ~ JS	Common factor	0.096069429	NA	NA	NA
6	SI ~ JS	Common factor	0.131226588	NA	NA	NA
7	SI ~ OC	Common factor	0.490013776	NA	NA	NA

Note that the standard errors, t -values, and p -values in the `summarize()` output are reported as NA. The reason is that a plain `csem()` call computes point estimates only. To conduct statistical inference, the standard errors of the estimates must be obtained by a resampling procedure such as the bootstrap, which is introduced in Section 4.

4 Assessment methods in cSEM

The next step involves assessing the model estimates, according to the guidelines described in Benitez et al. (2020). Before any results are interpreted, we first validate the estimation. Second we assess the model adequacy, with $SRMR$, d_{uls} and d_G . Third we subsequently examine the reliability of construct scores, with the Dijkstra–Henseler’s ρ_A . Fourth we check the factor loading estimates and significance. Fifth we assess construct validity, i.e. convergent and discriminant validity (Campbell & Fiske, 1959). Convergent validity is assed via the average variance extracted AVE . Discriminant validity is assed with the $HTMT$ and its improved version $HTMT2$. Sixth we check for multicollinearity, via VIF . Seventh we check the weights and loadings values and their significance.

Validation of the estimation

Before we interpret any results, we need to ensure the estimation is valid. This is done with the `verify()` function. This function gives the user details about the estimation. It gives information if convergence is achieved, all absolute standardized loading estimates are smaller or equal than one. It also checks of the construct variance covariance matrix is positive semi-definite, if all reliability estimates are smaller or equal than one and if the model-implied indicator variance covarariance matrix is positive semi-definite. It also checks if the construct and the model-implied indicator variance covariance matrix are semi positive definite and if all reliability estimates are smaller or equal than one. If a check fails, the results should not be interpreted and the user needs to customize the algorithm to ensure a valid estimation process. For that check Section ??.

```
# Check whether the estimation is admissible
verify(res_csem)
```

```
-----
Verify admissibility:
```

```
admissible
```

```
Details:
```

```
Code    Status    Description
```

1	ok	Convergence achieved
2	ok	All absolute standardized loading estimates <= 1
3	ok	Construct VCV is positive semi-definite
4	ok	All reliability estimates <= 1
5	ok	Model-implied indicator VCV is positive semi-definite

Testing the adequacy of the model

In this section we test the overall model fit. The function `testOMF()` is a bootstrap-based test for model fit originally proposed by ?. See also ? who first suggested the test in the context of PLS-PM.

```
# Bootstrap-based test of the overall model fit
set.seed(42)
fit_test <- testOMF(
  res_csem,
  .R = 1000,
  #.seed = 42,
  .handle_inadmissibles = "replace"
  #.Random.seed = 42
)
```

```
fit_test
```

```
----- Test for overall model fit based on Beran & Srivastava (1985) -----
```

Null hypothesis:

```
+-----+
|
|  H0: The model-implied indicator covariance matrix equals the
|  population indicator covariance matrix.
|
+-----+
```

Test statistic and critical value:

Distance measure	Test statistic	Critical value
		95%
dG	0.3253	0.3471
SRMR	0.0706	0.0500
dL	1.1505	0.5764
dML	1.8972	2.0046

Decision:

Distance measure	Significance level
	95%
dG	Do not reject

SRMR	reject
dL	reject
dML	Do not reject

Additional information:

Out of 2011 bootstrap replications 1000 are admissible.
See `?verify()` for what constitutes an inadmissible result.

The seed used was: -1809088912

By default `testOMF()` test the null hypothesis that the population indictator correlation matrix equals the population model-implied indicator correlation matrix. `testOMF()` uses different distance measures to assess the distance between the sample indicator correlation matrix and the estimated model-implied indicator correlation matrix, namely the geodesic distance (d_G), the squared Euclidean distance d_{ULS} and the root mean square residual $SRMR$.

Evaluating the reliability of the construct scores

The `assess()` function computes all relevant quality criteria:

```
#Compute all quality criteria
quality <- assess(res_csem)
```

```
Warning: The following warning occured in the calculateHTMT() function:
Intra-block and inter-block correlations between indicators must be either all-positive
or all-negative.
Warning: The following warning occured in the calculateHTMT() function:
Intra-block and inter-block correlations between indicators must be either all-positive
or all-negative.
Warning in log(x[[3]]): NaNs produced
```

```
#Show Dijkstras and Henselers rho A
quality$Reliability

$Cronbachs_alpha
      EP      AC      JS      OC      SI
0.8595716 0.8936315 0.8450310 0.8328395 0.8890480

$Joereskogs_rho
      EP      AC      JS      OC      SI
0.8607098 0.8918737 0.8417418 0.8343858 0.8901518

$`Dijkstra-Henselers_rho_A`
      EP      AC      JS      OC      SI
0.8717254 0.9105299 0.8568177 0.8878870 0.8989122
```

The results indicate that all constructs meet the commonly accepted thresholds. All constructs succeed the recommend threshold of 0.707 for Dijkstra-Henseler ρ_a , indicating reliable construct scores. Additionnaly `assess()` outputs alternative reliability estimates.

```
# Show cronbach alpha
quality$Reliability

$Cronbachs_alpha
      EP      AC      JS      OC      SI
0.8595716 0.8936315 0.8450310 0.8328395 0.8890480

$Joereskogs_rho
      EP      AC      JS      OC      SI
0.8607098 0.8918737 0.8417418 0.8343858 0.8901518

$`Dijkstra-Henselers_rho_A`
      EP      AC      JS      OC      SI
0.8717254 0.9105299 0.8568177 0.8878870 0.8989122
```

```
# Show rho_C
quality$RhoC
      EP      AC      JS      OC      SI
0.8607098 0.8918737 0.8417418 0.8343858 0.8901518

quality$RhoC_mm
      EP      AC      JS      OC      SI
0.8556731 0.8765202 0.8266506 0.8029492 0.8860215

quality$RhoC_weighted
      EP      AC      JS      OC      SI
0.8717254 0.9105299 0.8568177 0.8878870 0.8989122

quality$RhoC_weighted_mm
      EP      AC      JS      OC      SI
0.8717254 0.9105299 0.8568177 0.8878870 0.8989122
```

Evaluating indicator reliability

```
# Re-estimate with bootstrap resampling for inference
res_csem_boot <- csem(
  .data      = HBAT_SEM_rel,
  .model     = model,
  .resample_method = "bootstrap",
  .R         = 1000,
  .seed      = 42
)
```

```
# Loading estimates incl. bootstrap std. errors, t-values and p-values
summarize(res_csem_boot)$Estimates$Loading_estimates
```

	Name	Construct_type	Estimate	Std_err	t_stat	p_value
1	EP =~ EP1	Common factor	0.6341011	0.08825730	7.184687	6.736133e-13
2	EP =~ EP2	Common factor	0.8786819	0.06817644	12.888350	5.235172e-38
3	EP =~ EP3	Common factor	0.7678021	0.08865615	8.660449	4.699094e-18
4	EP =~ EP4	Common factor	0.8230691	0.06234019	13.202864	8.446117e-40
5	AC =~ AC1	Common factor	0.7691299	0.08208943	9.369415	7.293568e-21
6	AC =~ AC2	Common factor	0.6765259	0.09899325	6.834061	8.254389e-12
7	AC =~ AC3	Common factor	0.8215860	0.07771581	10.571671	4.032442e-26
8	AC =~ AC4	Common factor	0.9933858	0.04413543	22.507672	3.491295e-112
9	JS =~ JS1	Common factor	0.6248337	0.11846811	5.274278	1.332798e-07
10	JS =~ JS2	Common factor	0.6964587	0.10384121	6.706959	1.987220e-11
11	JS =~ JS3	Common factor	0.7182274	0.10374878	6.922755	4.429422e-12
12	JS =~ JS4	Common factor	0.6319045	0.12166320	5.193883	2.059521e-07
13	JS =~ JS5	Common factor	0.9004003	0.09311164	9.670115	4.039225e-22
14	OC =~ OC1	Common factor	0.4252105	0.06655371	6.388983	1.669924e-10
15	OC =~ OC2	Common factor	0.9574682	0.03086106	31.025128	2.470773e-211
16	OC =~ OC3	Common factor	0.6858765	0.05274373	13.003943	1.161946e-38
17	OC =~ OC4	Common factor	0.8565670	0.04700870	18.221458	3.487351e-74
18	SI =~ SI1	Common factor	0.8170099	0.04630598	17.643722	1.137185e-69
19	SI =~ SI2	Common factor	0.8705194	0.03913952	22.241445	1.364890e-109
20	SI =~ SI3	Common factor	0.6778349	0.05957767	11.377332	5.423357e-30
21	SI =~ SI4	Common factor	0.8959341	0.04423967	20.251826	3.423134e-91
	CI_percentile.95%L		CI_percentile.95%U			
1			0.4532620		0.7798607	
2			0.7214431		0.9805613	
3			0.5959517		0.9257479	
4			0.6938317		0.9348749	
5			0.6214292		0.9414872	
6			0.4665006		0.8574168	
7			0.6735746		0.9734165	
8			0.8318356		0.9978215	
9			0.3740575		0.8383272	
10			0.4843479		0.8895589	
11			0.4989930		0.8887790	
12			0.3733224		0.8466869	
13			0.6369939		0.9889399	
14			0.2851689		0.5408111	
15			0.8851819		0.9963862	
16			0.5873668		0.7951513	
17			0.7585917		0.9460804	
18			0.7268224		0.9095172	
19			0.7801786		0.9365719	
20			0.5648917		0.7977440	
21			0.8073558		0.9718306	

Evaluating convergent validity

Average variance extracted (AVE)

quality\$AVE

EP	AC	JS	OC	SI
0.6102822	0.6777668	0.5202693	0.5754208	0.6718668

The average variance extracted (AVE) for all constructs are above the recommended threshold of 0.5, supporting convergent validity.

Evaluating discriminant validity

To assess discriminant validity users can utilize the HTMT (Henseler, Ringle, & Sarstedt, 2015) which is calculated with the default options within the `assess()` function. There are several ways on testing for discriminant validity. One can compare the HTMT with a threshold, common values are 0.85 and 0.9 (Henseler et al., 2015). Or we can do inference by calculating the confidence interval bounds and comparing the upper limit with one. For that the `.inference` needs to be set on `TRUE`. If one is bigger than the upper limit then discriminant validity is established.

```
# Heterotrait-monotrait ratio of correlations (HTMT)
```

```
quality$HTMT
```

```
$htmts
```

	EP	AC	JS	OC	SI
EP	1.0000000	0.0000000	0.0000000	0.0000000	0
AC	0.2555939	1.0000000	0.0000000	0.0000000	0
JS	0.2437979	0.05239146	1.0000000	0.0000000	0
OC	0.4939933	0.27449892	0.2086788	1.0000000	0
SI	0.5672796	0.30936763	0.2313363	0.5005239	1

```
$quantiles
```

```
NULL
```

```
$nr_admissibles
```

```
NULL
```

```
quality_inf$HTMT
```

```
Error:
```

```
! object 'quality_inf' not found
```

Assessing multicollinearity among the indicators

```
# VIF values for Mode B (composite) weights
```

```
quality$VIF_modeB
```

Evaluating the weights

```
# Weight estimates incl. bootstrap std. errors, t-values and p-values
```

```
summarize(res_csem_boot)$Estimates$Weight_estimates
```

	Name	Construct_type	Estimate	Std_err	t_stat	p_value
1	EP <~ EP1	Common factor	0.2425256	0.03033059	7.996073	1.284500e-15
2	EP <~ EP2	Common factor	0.3360708	0.03417091	9.834998	7.956905e-23

3	EP <~ EP3	Common factor	0.2936625	0.02620371	11.206907	3.771439e-29
4	EP <~ EP4	Common factor	0.3148005	0.02512880	12.527478	5.281529e-36
5	AC <~ AC1	Common factor	0.2707114	0.02974091	9.102324	8.841910e-20
6	AC <~ AC2	Common factor	0.2381175	0.03264047	7.295162	2.982994e-13
7	AC <~ AC3	Common factor	0.2891744	0.02794699	10.347249	4.306835e-25
8	AC <~ AC4	Common factor	0.3496430	0.01889744	18.502136	1.984503e-76
9	JS <~ JS1	Common factor	0.2223363	0.04059423	5.477043	4.324929e-08
10	JS <~ JS2	Common factor	0.2478229	0.03710726	6.678555	2.413101e-11
11	JS <~ JS3	Common factor	0.2555689	0.03888206	6.572926	4.933583e-11
12	JS <~ JS4	Common factor	0.2248524	0.04381154	5.132264	2.862779e-07
13	JS <~ JS5	Common factor	0.3203920	0.03598888	8.902527	5.458932e-19
14	OC <~ OC1	Common factor	0.1740754	0.02457108	7.084563	1.394839e-12
15	OC <~ OC2	Common factor	0.3919744	0.01869390	20.968041	1.284457e-97
16	OC <~ OC3	Common factor	0.2807885	0.02061076	13.623392	2.907433e-42
17	OC <~ OC4	Common factor	0.3506668	0.01644880	21.318690	7.616096e-101
18	SI <~ SI1	Common factor	0.2882324	0.01534997	18.777390	1.156254e-78
19	SI <~ SI2	Common factor	0.3071099	0.01550302	19.809683	2.456267e-87
20	SI <~ SI3	Common factor	0.2391329	0.01876727	12.742019	3.453427e-37
21	SI <~ SI4	Common factor	0.3160760	0.01634721	19.335168	2.717827e-83
CI_percentile.95%L CI_percentile.95%U						
1			0.1828087		0.2999783	
2			0.2725098		0.4008700	
3			0.2414227		0.3459511	
4			0.2697347		0.3683087	
5			0.2183269		0.3399151	
6			0.1723162		0.2988541	
7			0.2386440		0.3504807	
8			0.2919151		0.3651913	
9			0.1423157		0.3013401	
10			0.1794724		0.3181615	
11			0.1792914		0.3324710	
12			0.1395502		0.3054149	
13			0.2341224		0.3685690	
14			0.1229501		0.2179109	
15			0.3530282		0.4250516	
16			0.2467718		0.3260126	
17			0.3185797		0.3842689	
18			0.2590976		0.3185720	
19			0.2763323		0.3387036	
20			0.2055149		0.2781864	
21			0.2820537		0.3429069	

References

- Barba, L. A. (2022). Defining the role of open source software in research reproducibility. *Computer*, 55(8), 40–48. doi: 10.1109/mc.2022.3177133
- Benitez, J., Henseler, J., Castillo, A., & Schuberth, F. (2020). How to perform and report an impactful analysis using partial least squares: Guidelines for confirmatory and explanatory IS research. *Information & Management*, 57(2), 103168. doi: 10.1016/j.im.2019.05.003
- Browne, M. W. (1974). Generalized least squares estimators in the analysis of covariance structures. *South African Statistical Journal*, 8(1), 1–24. doi: 10.1002/j.2333-8504.1973.tb00197.x
- Campbell, D. T., & Fiske, D. W. (1959). Convergent and discriminant validation by the multitrait-multimethod matrix. , 56(2), 81–105. (Place: US) doi: 10.1037/h0046016
- Cheah, J.-H., Magno, F., & Cassia, F. (2024). Reviewing the SmartPLS 4 software: the latest features and enhancements. *Journal of Marketing Analytics*, 12, 97–102. doi: 10.1057/s41270-023-00266-y
- Cheah, J.-H., Sarstedt, M., Hair, J. F., & Ringle, C. M. (2026). Consistent partial least squares structural equation modeling using smartpls. *Structural Equation Modeling: A Multidisciplinary Journal*, 1–14. doi: 10.1080/10705511.2026.2633754
- Cook, R. D., & Forzani, L. (2023). On the role of partial least squares in path analysis for the social sciences. *Journal of Business Research*, 167, 114132. doi: 10.1016/j.jbusres.2023.114132
- Devlieger, I., & Rosseel, Y. (2017). Factor score path analysis. *Methodology*, 13, 31–38.
- Dijkstra, T. K. (1985). *Latent variables in linear stochastic models: Reflections on "maximum likelihood" and "partial least squares" methods* (Vol. 1). Amsterdam: Sociometric Research Foundation.
- Dijkstra, T. K. (2013). *A note on how to make pls consistent* (Tech. Rep.). working paper, University of Groningen, Groningen, The Netherlands. Retrieved from <http://www.rug.nl/staff/tkdijkstra/how-to-make-pls-consistent.pdf>
- Dijkstra, T. K., & Henseler, J. (2015a). Consistent and asymptotically normal PLS estimators for linear structural equations. *Computational Statistics & Data Analysis*, 81, 10–23. doi: 10.1016/j.csda.2014.07.008
- Dijkstra, T. K., & Henseler, J. (2015b). Consistent partial least squares path modeling. *MIS Quarterly*, 39(2), 297–316. doi: 10.25300/MISQ/2015/39.2.02
- Dijkstra, T. K., & Schermelleh-Engel, K. (2014). Consistent partial least squares for nonlinear structural equation models. *Psychometrika*, 79(4), 585–604. doi: 10.1007/s11336-013-9370-0
- Evermann, J., & Rönkkö, M. (2023). Recent developments in PLS. *Communications of the Association for Information Systems*, 52, 663–667. doi: 10.17705/1CAIS.05229
- Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2019). *Multivariate data analysis* (8th ed.). Cengage.
- Hair, J. F., Ringle, C. M., & Sarstedt, M. (2011). PLS-SEM: Indeed a silver bullet. *Journal of Marketing Theory and Practice*, 19(2), 139–152. doi: 10.2753/MTP1069-6679190202
- Henseler, J. (2025). Adanco 2.4 [Computer software manual]. Enschede.
- Henseler, J., Ringle, C. M., & Sarstedt, M. (2015). A new criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy of Marketing Science*, 43(1), 115–135. doi: 10.1007/s11747-014-0403-8
- Hui, B. S., & Wold, H. (1982). Consistency and consistency at large of partial least squares estimates. In K. G. Jöreskog & H. Wold (Eds.), *Systems under indirect observation: Causality, structure, prediction part ii* (pp. 119–130). Amsterdam: North-Holland.
- Humble, J. (2024). plspm: A library implementing partial least squares path modeling [Com-

- puter software manual]. Retrieved from <https://plspm.readthedocs.io/> (Python package version 0.5.7)
- Hwang, H., Cho, G., Jung, K., Falk, C., Flake, J., Jin, M. J., & Lee, S. H. (2021). An approach to structural equation modeling with both factors and components: Integrated generalized structured component analysis. *Psychological Methods*, 26(3), 273–294. doi: 10.1037/met0000336
- Hwang, H., & Takane, Y. (2004). Generalized structured component analysis. *Psychometrika*, 69(1), 81–99. doi: 10.1007/BF02295841
- Ince, D. C., Hatton, L., & Graham-Cumming, J. (2012). The case for open computer programs. *Nature*, 482(7386), 485–488. doi: 10.1038/nature10836
- Jöreskog, K. G. (1970). A general method for estimating a linear structural equation system. *ETS Research Bulletin Series*, 1970(2), i–41. doi: 10.1002/j.2333-8504.1970.tb00783.x
- Jung, S., & Park, J. (2018). Consistent partial least squares path modeling via regularization. *Frontiers in Psychology*, 9. doi: 10.3389/fpsyg.2018.00174
- Kline, R. B. (2023). *Principles and practice of structural equation modeling* (5th ed.). New York: The Guilford Press.
- Kock, N. (2025). *WarpPLS user manual: Version 8.0*. Laredo, TX: ScriptWarp Systems.
- Marcoulides, G. A., & Schumacker, R. E. (Eds.). (1996). *Advanced structural equation modeling: Issues and techniques*. New York: Psychology Press.
- Morin, A., Urban, J., Adams, P. D., Foster, I., Sali, A., Baker, D., & Sliz, P. (2012). Shining light into black boxes. *Science*, 336(6078), 159–160. doi: 10.1126/science.1218263
- Müller, T., Schuberth, F., & Henseler, J. (2018). PLS path modeling - a confirmatory approach to study tourism technology and tourist behavior. *Journal of Hospitality and Tourism Technology*, 9(3), 249–266. doi: 10.1108/JHTT-09-2017-0106
- Python Software Foundation. (2024). Python language reference, version 3.12 [Computer software manual]. Retrieved from <https://www.python.org>
- R Core Team. (2025). R: A language and environment for statistical computing [Computer software manual]. Vienna, Austria. Retrieved from <https://www.R-project.org/>
- Rademaker, M. E., & Schuberth, F. (2020). csem: Composite-based structural equation modeling [Computer software manual]. Retrieved from <https://floschuberth.github.io/cSEM/> (Package version: 0.6.1.9000)
- Ray, S., Danks, N. P., & Calero Valdez, A. (2026). seminr: Building and estimating structural equation models [Computer software manual]. Retrieved from <https://CRAN.R-project.org/package=seminr> (R package version 2.4.2) doi: 10.32614/CRAN.package.seminr
- Ringle, C. M., Sarstedt, M., Mitchell, R., & Gudergan, S. P. (2020, jan). Partial least squares structural equation modeling in HRM research. *The International Journal of Human Resource Management*, 31(12), 1617–1643. doi: 10.1080/09585192.2017.1416655
- Ringle, C. M., Wende, S., & Becker, J.-M. (2024). *SmartPLS 4*. Boenningstedt: SmartPLS GmbH. Retrieved from <https://www.smartpls.com>
- Rönkkö, M. (2022). matrixpls: Matrix-based partial least squares estimation [Computer software manual]. Retrieved from <https://github.com/mronkko/matrixpls> (R package version 1.0.15)
- Rönkkö, M., & Evermann, J. (2013). A critical examination of common beliefs about partial least squares path modeling. *Organizational Research Methods*, 16(13), 425–448. doi: 10.1177/1094428112474693
- Sanchez, G., Trinchera, L., Russolillo, G., & Bertrand, F. (2025). Partial least squares path modeli (PLS-PM, Frederic) [Computer software manual]. Retrieved from <https://CRAN.R-project.org/package=plspm> (R package version 0.6.0) doi: 10.32614/CRAN.package.plspm
- Sarstedt, M., Hair, J. F., Pick, M., Liengard, B. D., Radomir, L., & Ringle, C. M. (2022).

- Progress in partial least squares structural equation modeling use in marketing research in the last decade. *Psychology & Marketing*, 39, 1035–1064. doi: 10.1002/mar.21640
- Schamberger, T., Schubert, F., Henseler, J., & Dijkstra, T. K. (2020). Robust partial least squares path modeling. *Behaviormetrika*, 47, 307–334. doi: 10.1007/s41237-019-00088-2
- Schubert, F., Henseler, J., & Dijkstra, T. K. (2018). Partial least squares path modeling using ordinal categorical indicators. *Quality & Quantity*, 52(1), 9–35. doi: 10.1007/s11135-016-0401-7
- Schubert, F., Rosseel, Y., Rönkkö, M., Trinchera, L., Kline, R. B., & Henseler, J. (2023). Structural parameters under partial least squares and covariance-based structural equation modeling: A comment on Yuan and Deng (2021). *Structural Equation Modeling: A Multidisciplinary Journal*, 30(3), 339–345. doi: 10.1080/10705511.2022.2134140
- Schubert, F., Zaza, S., & Henseler, J. (2023). Partial least squares is an estimator for structural equation models: A comment on Evermann and Rönkkö (2021). *Communications of the Association for Information Systems*, 52, 711–729. doi: 10.17705/1CAIS.05232
- Skrondal, A., & Laake, P. (2001). Regression among factor scores. *Psychometrika*, 66(4), 563–575. doi: 10.1007/BF02296196
- UNESCO. (2021). *UNESCO recommendation on open science*. Paris: UNESCO. doi: 10.54677/MNMH8546
- Venturini, S., Mehmetoglu, M., & Latan, H. (2023). Software packages for partial least squares structural equation modeling: An updated review. In *Partial least squares path modeling* (pp. 113–152). Springer International Publishing. doi: 10.1007/978-3-031-37772-3_5
- Wold, H. (1982). Soft modeling: The basic design and some extensions. In K. G. Jöreskog & H. Wold (Eds.), *Systems under indirect observation: Causality, structure, prediction part ii* (pp. 1–54). Amsterdam: North-Holland.